

ACTIVE PCA SYSTEMS

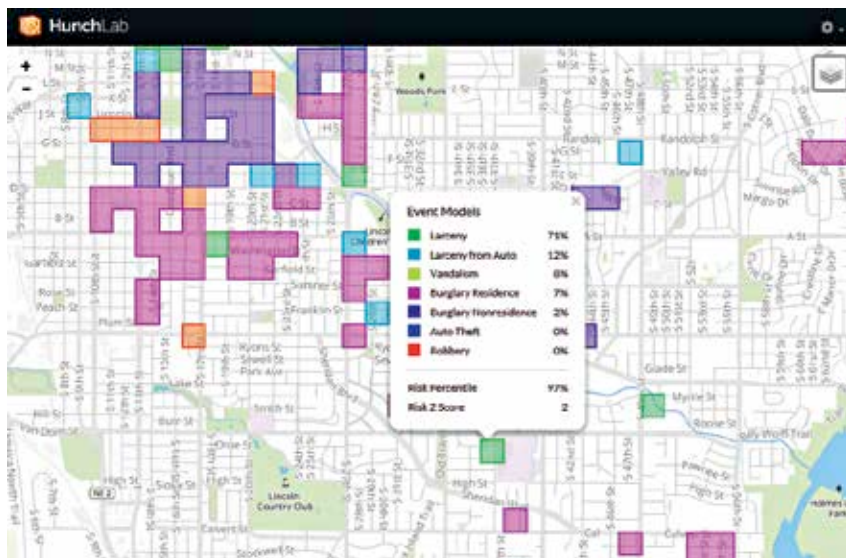
As mentioned earlier, quite a few systems are already in place and have been aiding law enforcement agencies. Here are some of them.

Hitachi Visualization Predictive Crime Analytics

Hitachi's offering uses data from the 911 dispatch system, license plate readers along the roads, gunshot sensors and other public safety systems to produce a heat map which indicates hot spots in the city with a higher probability of crime occurring. It aims to provide policing agencies with real-time insight so that these agencies can be better prepared for an incident before it even occurs – prevention is better than cure when it comes to crime, after all. It's a hybrid cloud-based system that can be implemented on any of the popular cloud service providers.

HunchLab

Made by a startup in Philadelphia by the name of Azavea, HunchLab is a platform that helps police balance resource allocation based of existing circumstances, such as the weather, crimes committed in the past few days, GPS data, etc., to generate a visualization of hotspots around a city. And by implementing tried-and-tested crime mitigation tactics, HunchLab can tell you where to deploy police officers and what situations they should expect. Accordingly, the police will be better equipped to handle the situations, and there's less of a surprise element.



HunchLab real time Risk Scoring

IBM SPSS Crime Prediction and Prevention

IBM isn't new to the world of AI, and rightly so, their system seems to be a lot more advanced. This platform monitors violent criminals to generate heat maps to point out which areas they frequent, and then it compares local incidents of crime with regional and national gang activities in order to extrapolate trends. These trends help the system identify criminals based on similarities related to previous crimes. Also, since it looks at previous crimes, it can identify and shortlist criminals whose modus operandi match up with observed activity. Add to this, the fact that it also looks at circumstantial data such as the weather, social events, holidays, etc. to see

which circumstances make for higher probability of criminal activity.

Domain Awareness System (DAS)

Conceptualised after the 9/11 terrorism attacks. DAS is a surveillance system built by the city of New York and Microsoft to monitor the city. It obtains real-time video from over six thousand street cameras in New York and even data from radiation sensors built into patrolling boats, cars, trucks and even the gun belts that each police officer carries. The sensors are so sensitive that they are also triggered by someone who has just had radiation therapy for diseases such as cancer. The sheer amount of metadata it generates and monitors, allows the system to track down individuals with simple keywords such as "wearing a red shirt".



Hitachi's system in action

THE FUTURE

Given how these systems are slowly evolving to identify trigger events which might flare up into an act of crime, one can always envision such platforms morphing into an even more perverse form. These systems are already monitoring schools so imagining a situation where children are identified early on as troublemakers would allow society to help them and mitigate crime. An even more perverse technology would result in genetic markers being observed to design babies who'll never walk down the path of crime – if we allow it, that is. **[1]**